

Adaptive Risk-Aware Autonomy for Robotic Exploration of Uncertain Extraterrestrial Terrain

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Abstract

A planetary rover cannot be driven in real time from Earth, so it must decide for itself which ground is safe enough to cross. Those decisions rest on a terrain model built before launch, from whatever experience was available at the time. This work asks what happens when that model is wrong — when the robot arrives somewhere whose ground does not behave the way its model says — and whether a robot that revises its model from its own driving survives better than one that does not.

We build a controlled 2.5-D autonomy testbed in which a single solar-powered rover must visit science targets and return to its lander before its energy runs out, on terrain where wheel slip is stochastic, sensing is noisy and range-limited, and sustained slip ends the mission by embedding the vehicle. Three planners are compared on identical terrain: distance-only A*, a fixed risk-aware planner, and an otherwise identical planner whose mobility model is updated online from measured slip. The three differ only in their objective and in whether that objective’s inputs are revised, so a measured difference is attributable to the decision rule rather than to a different search implementation.

The central experiment is a domain shift. Every planner carries a mobility prior calibrated on lunar terrain and is then evaluated on Martian terrain, where the same terrain-class vocabulary carries substantially different slip statistics. Across 750 missions on frozen, held-out seeds, we report the effect of online adaptation on mission success and science return, and the generalization gap between in-distribution and out-of-distribution performance.

The paper reports two studies. *Study 1* is complete: a confirmatory run on engine profile v1, reported with its pre-specified tests. An audit after it found simulator and learner defects that change what several of its conditions measured; those findings, their fixes, and a calibration audit of the terrain learner on development data are reported as such. *Study 2* is planned and not yet executed: its question, hypotheses, sample-size analysis and seed protocol are given, and it reports no results.

This is a simulation study of decision-making, not a validated model of any flight vehicle or landing site. Section 10 states plainly what the model does and does not represent.

1 Introduction

Round-trip light time to Mars runs from roughly eight to forty minutes, and contact windows are intermittent. A surface robot therefore cannot be joysticked around a hazard; by the time an operator sees a problem, the problem has already happened. Autonomy on planetary rovers exists for this reason, and the quality of that autonomy is measured by what the robot does when nobody can help it.

The failure mode that actually strands planetary rovers is not driving off a cliff. It is sinking. A rover crossing ground that looks benign can lose traction in loose material, dig itself in, and become permanently immobile. NASA’s *Spirit* ended its driving mission this way in 2009, embedded in soft soil [7]. What makes this failure mode interesting for autonomy research is that it is invisible to geometry: the ground that traps a rover is often flat. Detecting it requires a model of how terrain *behaves*, not how it *looks*, and the only direct measurement of behaviour is slip observed while driving on it.

That creates an asymmetry this paper is built around. Geometry — slope, roughness, obstacles — can be sensed at a distance from imagery. Mobility can only be measured by committing the vehicle to the ground. A robot therefore plans long routes using a terrain model it cannot verify in advance, and the model it starts with was calibrated somewhere else.

Research question. Does a planetary robot that continuously updates its terrain-mobility model from its own driving make safer and more productive decisions than one using a fixed model, when the terrain it encounters does not match the model it was given?

Study 1 hypotheses. Stated before the Study 1 confirmatory data were generated (see docs/PREREGISTRATION).

- H1** On in-distribution terrain, where the prior is already correct, adaptation neither helps nor harms: the adaptive and fixed planners perform equivalently.
- H2** On out-of-distribution terrain, the adaptive planner returns a higher paired science fraction than the fixed planner.
- H3** Under increased slip dispersion and injected faults, the adaptive planner achieves a higher mission-success rate and fewer severe-slip events.
- H4** Intervention delay reduces productivity for all planners, and most for those that request more interventions.

What is and is not claimed as novel. Risk-aware planetary path planning, learned traversability, uncertainty-aware terrain models and energy-constrained planning all exist in the literature and are cited in Section 2. This work does not claim to invent any of them. What it contributes is a controlled measurement: holding the search algorithm, the cost function, the mission logic and the terrain fixed, and changing *only* whether the mobility model is revised online, in order to isolate what that single mechanism is worth under a realistic domain shift.

Structure. Sections 3–4 describe the simulator and the autonomy stack. Section 5 reports Study 1 as run. Section 7 reports what was found after it — simulator defects, their effect on Study 1, and a calibration audit of the terrain learner — and Section 8 describes Study 2, whose plan is frozen and which has not been run.

2 Related work

Slip has long been treated as the operative traversability signal for Mars rovers rather than slope alone; Angelova et al. [1] learned to predict slip from terrain appearance, and Helmick et al. [4] built terrain-adaptive navigation around slip-aware models. More recent work makes the uncertainty explicit rather than producing a single traversability score: Endo et al. [3] fuse probabilistic

traversability predictions for risk-aware planning on heterogeneous terrain, and extend this with test-time adaptation [2] so that a traversability model can be corrected during deployment. Sanchez-Ibanez et al. [8] address a complementary risk, planning under localization uncertainty for space exploration rovers.

The distinction this paper draws — between uncertainty that more driving can reduce and variability that it cannot — follows the same motivation as that line of work. Our contribution is not a better traversability model but a controlled comparison of what online correction of such a model is worth when the deployment environment differs from the calibration environment.

3 Simulation model

3.1 Scope

The simulator is a controlled testbed for comparing decision rules. It is not a validated dynamics model of a specific rover, and its absolute outputs are properties of the model rather than predictions of on-surface performance. Its purpose is to expose different planners to *identical* stochastic worlds so that differences between them are attributable to their decisions.

3.2 Terrain

Terrain is a $N \times N$ grid with an elevation field $z(x)$, from which slope $\theta(x)$ is the gradient magnitude in degrees. Each cell carries a terrain class $k(x)$ from a five-member vocabulary shared by both bodies: smooth regolith, rocky, loose fines, bedrock, and rim talus. Each cell also carries a roughness value, an illumination fraction $I(x) \in [0, 1]$, and a hazard flag.

Crucially, the class *vocabulary* is shared across bodies while the class *parameters* are not. Each body assigns every class a true mean slip μ_k , dispersion σ_k , and locomotion multiplier m_k . This is the mechanism of the domain shift: a robot arriving at a new body recognizes the same terrain categories but the statistics behind them have changed.

3.3 Mobility and embedding

When the robot commands a move into cell x , the realized slip fraction is

$$s \sim \mathcal{N}\left(\mu_{k(x)} + \beta \theta(x) + b, \sigma_{k(x)}^2\right), \quad s \in [0, 0.995], \quad (1)$$

with $\beta = 0.01$ per degree, representing traction lost on grades, and b a bias introduced by wheel-damage faults. A step is *severe* when $s \geq \tau$ with $\tau = 0.80$: the wheels turn but the robot does not advance. After $L = 3$ consecutive severe steps the robot is permanently embedded and the mission ends. This is the model’s representation of the Spirit-style loss mode, and it is the only failure with no recovery.

3.4 Energy

Energy to traverse a cell is

$$E(x) = \frac{E_0 d m_{k(x)} (1 + \gamma \theta(x))}{\max(1 - s, \epsilon)} \cdot \frac{g}{g_{\text{ref}}} \cdot \frac{1}{\eta_m}, \quad (2)$$

with d the distance in cells, $\gamma = 0.035$ per degree, $\epsilon = 0.08$, g the body’s gravity against a lunar reference $g_{\text{ref}} = 1.62 \text{ m/s}^2$, and η_m a motor-efficiency factor degraded by faults. The slip term sits

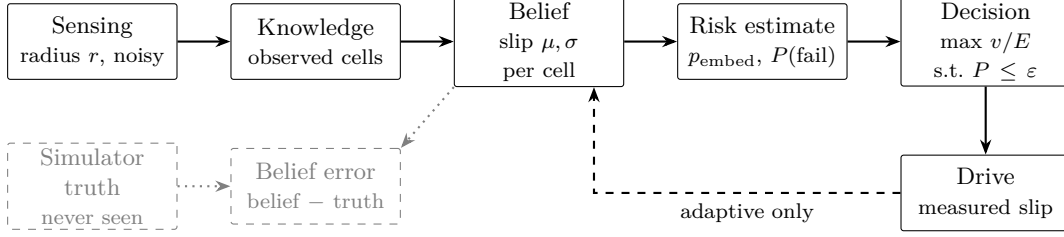


Figure 1: Information flow in the autonomy stack. Solid boxes are what the rover computes and acts on. Dashed grey boxes are evaluation quantities available to the study but not to the rover; belief error explains decisions after the fact and is not an input to them. The dashed arrow (measured slip revising belief) exists for the adaptive planner only.

in the denominator because slipping wheels still consume power while delivering less ground: at slip s , covering the same distance costs roughly $1/(1-s)$ times as much.

Equation 2 is the reason a mobility error is also an energy error, and it is the mechanism behind the results in Section 6. A robot that underestimates slip underestimates the cost of every route it considers.

Solar income per timestep is $\rho I(x) \eta_s$, with ρ the nominal harvest rate and η_s an efficiency degraded by dust faults. Battery capacity is scaled by g/g_{ref} , representing a vehicle whose power system is sized for the body it was built for; what transfers between bodies in this study is the autonomy, not the hardware.

3.5 Sensing

The robot never reads ground truth. Within a sensing radius it receives noisy estimates of slope, roughness, illumination, terrain class and hazard status, with noise growing linearly in range. Terrain class is misclassified with a range-dependent probability. Slip, by contrast, is measured directly and without class ambiguity — but only for the single cell just driven.

4 Autonomy stack

Figure 1 shows how information moves through the stack. The rover senses a disc around itself, which grows its map of observed cells; its belief about slip combines those observations with the class prior and, for the adaptive planner only, with slip measured while driving; risk is computed from that belief; and the mission manager chooses on risk. Belief *error*, the difference between belief and simulator truth, appears in the figure only as an evaluation branch: the rover never has truth, so error is something this study measures about the rover, not something the rover uses.

4.1 World model

The robot maintains a belief, not the terrain. Per cell it stores the most recent close-range geometric estimate. Per terrain class it stores a posterior over mean slip, updated from driven experience by a Normal–Normal conjugate model: with prior mean m_0 , prior variance t_0^2 , observation variance σ_o^2 and n measurements of sample mean $\bar{s}$,

$$\lambda = \frac{1}{t_0^2} + \frac{n}{\sigma_o^2}, \quad m_n = \frac{1}{\lambda} \left(\frac{m_0}{t_0^2} + \frac{n\bar{s}}{\sigma_o^2} \right), \quad t_n^2 = \frac{1}{\lambda}. \quad (3)$$

The posterior is recomputed from the prior and the full sample each time, so it does not depend on update order.

Generalizing to unseen ground. Routes are composed mostly of cells the robot has never observed, so what it believes about *unknown* terrain determines almost every plan. For an unobserved cell the class is unknown, and the expected slip is the current class beliefs marginalized over the class mix π_k actually encountered so far:

$$\hat{s}_{\text{unk}} = \sum_k \pi_k m_n^{(k)}. \quad (4)$$

This is what allows learning to transfer: measuring that the local drift sand is treacherous raises the robot’s estimate for ground it has not yet reached. A fixed constant in place of Equation 4 would confine every lesson to cells already visited and, because routes are mostly unvisited cells, would prevent adaptation from affecting any planning decision at all.

Two kinds of uncertainty. For unobserved cells the epistemic term combines the spread *between* class means with the uncertainty *within* each:

$$\sigma_{\text{epi}}^2 = \underbrace{\sum_k \pi_k (m_n^{(k)} - \hat{s}_{\text{unk}})^2}_{\text{between classes}} + \underbrace{\sum_k \pi_k t_n^{2(k)}}_{\text{within class}}, \quad \sigma_{\text{tot}}^2 = \sigma_{\text{epi}}^2 + \sigma_{\text{ale}}^2. \quad (5)$$

The separation matters: σ_{epi} is reducible by driving, whereas σ_{ale} is the terrain’s own variability and is not. A planner that cannot tell them apart cannot distinguish ground that is dangerous from ground that is merely unvisited.

4.2 Risk

Under the belief that slip at x is $\mathcal{N}(\hat{s}, \sigma_{\text{tot}}^2)$, the probability of a severe step and of embedding are

$$p_{\text{sev}}(x) = 1 - \Phi\left(\frac{\tau - \hat{s}(x)}{\sigma_{\text{tot}}(x)}\right), \quad p_{\text{emb}}(x) = p_{\text{sev}}(x)^L. \quad (6)$$

Treating consecutive steps as independent *understates* risk in homogeneous terrain, where severe slip is spatially correlated and runs are more likely than independence implies. The approximation is applied identically to every planner and so cannot favour one.

A believed hazard is not by itself mission-ending: the robot’s onboard geometric check refuses such a command and replans. It enters the risk only through the residual probability $\kappa = 0.005$ that the check misses it:

$$R(x) = 1 - (1 - p_{\text{emb}}(x))(1 - \kappa p_{\text{haz}}(x)), \quad R(\pi) = 1 - \prod_{x \in \pi} (1 - R(x)). \quad (7)$$

Energy risk is evaluated against the usable budget, treating required energy as normal with variance propagated from slip uncertainty by linearizing Equation 2 about $\hat{s}$:

$$\sigma_E^2 = \sum_{x \in \pi} \left(\frac{E(x)}{1 - \hat{s}(x)} \right)^2 \sigma_{\text{tot}}^2(x), \quad P_{\text{energy}} = \Pr[E_{\text{req}} \geq B + S - R_{\text{res}}], \quad (8)$$

with B the charge in hand, S the solar income expected over the route, and R_{res} a reserve held back as a fraction of capacity. Total mission failure probability composes the two causes.

4.3 Mission logic

A mission is: start at the lander, visit science targets, return before the energy runs out. The manager pursues the reachable target with the best value-to-cost ratio whose round trip satisfies the risk budget $P_{\text{fail}} \leq \varepsilon$, and otherwise returns home. Missions are flown as sorties — a target that does not fit on a low battery is reconsidered after recharging, and written off only when the robot is home, recharged, and still cannot reach it.

When no route is believed to exist, the robot must ask mission control. It idles for the communication delay before receiving a relaxed instruction, and each such episode is counted as a human intervention. An autonomy system that succeeds only by asking for help frequently has not really succeeded, so intervention count is reported alongside the outcome measures.

4.4 The three planners

All three run the same A* implementation over the same believed world and differ only in the cost assigned to entering a cell.

Distance-only A*

$J = d$. The control with no terrain model at all.

Fixed risk-aware A*

$J = w_d d + w_e E(x) + w_r R(x) + w_u \sigma_{\text{epi}}(x)$, consulting the world model but never revising it.

Adaptive risk-aware A*

Identical objective and identical weights; the single difference is that measured slip is folded into the class belief via Equation 3, so future plans use corrected estimates.

The adaptive planner is implemented as a subclass of the fixed planner that overrides one method. The experimental treatment is therefore a single switch, which is what makes the contrast interpretable.

5 Study 1: design

Study 1 ran on engine profile *v1*. Its committed results are locked: a SHA-256 of every artifact is recorded in `data/results/v1_LOCK.json`, a test fails if any of them changes, and all 750 missions were re-run and reproduced exactly, on the macOS/Apple-Silicon environment that generated them, before the lock (on Linux the conclusions are unchanged but not every mission is identical; Section 11). Defects found later (Section 7) are reported, not repaired, in Study 1’s results.

5.1 Seed protocol

Terrain seeds are partitioned into disjoint integer ranges — train, validation, test and out-of-distribution — frozen to a checksummed file before any result was generated. Priors are calibrated on train seeds only. All tuning was performed on validation seeds. Test and OOD seeds informed no design decision.

Eight seeds consumed by an earlier engineering pilot are permanently quarantined and excluded from every confirmatory result; a regression test asserts they cannot reappear. The confirmatory run used 50 seeds per condition from split checksum `c9346c400ce5` at commit `a5e5081728c8`.

5.2 Design

3 planners $\times$ 5 conditions $\times$ 50 seeds = 750 missions. Every planner is evaluated on identical terrain within a condition, making this a randomized block design in which terrain seed is the block.

Conditions: Moon in-distribution; Mars out-of-distribution; Mars with $1.5\times$ slip dispersion; Mars with injected faults; and Mars with a 20-step intervention delay. Every condition uses a lunar-calibrated prior, so the Martian conditions are genuine domain shifts.

5.3 Analysis

Because planners share terrain within a condition, all contrasts are paired. Science fraction uses a paired t -test with a Wilcoxon signed-rank companion, Cohen’s d_z , and a 95% confidence interval. Mission success is binary and paired, so it uses McNemar’s exact test [6], which conditions on the seeds where the two planners actually disagreed. Holm–Bonferroni correction [5] is applied across the prespecified primary family; secondary contrasts against the distance-only control are corrected separately and never pooled with it.

We also report the generalization gap $G = M_{\text{in}} - M_{\text{out}}$ per planner, a between-condition descriptive quantity reported without a paired test because the conditions draw from different seed splits.

6 Study 1: results

All 750 confirmatory missions completed without software error and none were excluded. Results are reported for every outcome regardless of significance, as the analysis plan requires.

6.1 In-distribution: adaptation is neutral (H1 supported)

On lunar terrain, where the prior is already correct, the adaptive and fixed risk-aware planners are statistically indistinguishable: success 0.92 versus 0.90 ($\Delta = +0.020$, $p_{\text{Holm}} = 1.000$), differing on only five of fifty seeds. This is what H1 predicted: when there is nothing to learn, learning costs nothing.

Both risk-aware planners clearly beat the distance-only control here (0.66 success, 0.659 science fraction), confirming that modelling terrain risk is worth doing when the model is right.

6.2 Out-of-distribution: H2 is not supported

H2 predicted that adaptation would raise science fraction on Martian terrain. It did not. In the baseline Martian condition the paired difference in science fraction was -0.015 ($p_{\text{Holm}} = 1.000$) — a point estimate in the *opposite* direction to the hypothesis — and mission success was exactly tied, with each planner winning five of the ten seeds on which they differed.

We report this as a failure of the stated hypothesis rather than reframing it. The validation-set estimate that motivated H2 (a success advantage of $+0.125$ to $+0.167$) did not replicate on held-out seeds. That divergence is itself the argument for holding seeds back: a tuning-set effect of that size vanished entirely under confirmatory evaluation.

6.3 Degraded conditions: H3 not supported as stated

The one contrast surviving Holm correction across the ten-member primary family is mission success in the condition labelled “hardware faults”, where the adaptive planner succeeded more often than

the fixed one (0.46 against 0.26, $\Delta = +0.200$, $p_{\text{Holm}} = 0.020$), with all ten discordant seeds favouring adaptation.

This cannot be read as an effect of faults. A post-hoc audit, which was not part of the analysis plan, re-ran every mission in this condition. The re-runs reproduced 100% of the committed rows exactly. Faults are scheduled uniformly across the 800-step horizon, but the median mission in this condition ended after 91 steps, so most scheduled faults never fired. A fault actually occurred in only 16% of adaptive missions and 26% of fixed ones. Of the 10 discordant seeds behind the significant result, a fault fired in both missions on 1, and in neither on 5.

So this condition mostly measured Martian missions without faults. It differs from the plain Martian condition, where the same contrast was exactly zero, chiefly through a shifted random stream (Section 10). The most defensible reading is that this is sampling variation that survived correction. We therefore do not claim support for H3. The other two degraded conditions moved in the same direction without reaching corrected significance: +0.140 under elevated slip dispersion ($p_{\text{Holm}} = 0.461$) and +0.160 under intervention delay ($p_{\text{Holm}} = 0.461$).

6.4 A consistent, unconfirmed sign pattern

Descriptively, in every Martian condition the adaptive planner’s success rate is at least the fixed planner’s and its science fraction is slightly lower, and it ends fewer missions in energy exhaustion. None of the science-fraction differences survive correction, and the pattern was not a pre-specified test.

A mechanism is plausible. Locomotion cost carries a $1/(1 - s)$ term (Equation 2), so a robot that raises its slip estimate raises its estimate of every route’s cost and turns back sooner. That trade would save missions and cost science. But it is a hypothesis, not a finding, and a further audit complicates it. The adaptive learner turned out to be badly overconfident. On 40 Martian validation missions, its 95% intervals for class-mean slip contained the true value only 0.20 of the time, against 0.95 for a calibrated model. Its median error was 0.060, while it reported a median uncertainty of just 0.0089. The cause is that each slip reading is treated as near-exact (Section 10). With the per-reading variance set to the terrain’s own dispersion, coverage rises only to 0.52 and median error falls to 0.044. The rest of the miscalibration comes from misclassification: the rover folds each reading into the class it *believes* it drove on, and some of those beliefs are wrong. So the method tested here adapts, but it adapts by over-reacting, and its effects should not be read as the effects of a well-calibrated learner.

6.5 An unanticipated result: the uninformed planner survives

None of the hypotheses anticipated the clearest pattern in the Martian data. Distance-only A*, which models no terrain risk whatsoever, achieved the *highest* mission-success rate of the three planners in three of the four Martian conditions — 0.48 against 0.34 and 0.34 in the baseline OOD condition.

It also returned the least science (0.101 against 0.173 and 0.158). The interpretation we favour, which the intervention counts support, is that risk-averse routing on hostile terrain is expensive: the risk-aware planners detour around suspect ground, and on Mars almost all ground is suspect, so they travel further, request far more ground interventions (102.8 per mission against 57.5 for distance-only), and more often run the battery down before getting home. The short, blunt route is more dangerous per metre but involves fewer metres.

This is a genuine limitation of the risk formulation rather than a curiosity. A planner that treats caution as free will over-pay for it when the whole map is dangerous.

6.6 Generalization gap

The generalization gap G is largest for the adaptive planner (0.670) and smallest for distance-only A* (0.558). This ranking should not be read as adaptation generalizing worst. G is a difference, and the adaptive planner starts from the highest in-distribution score, so the metric mechanically rewards planners that are poor at home. Reported for completeness, it is uninformative for ranking here.

Table 1: Outcomes by condition and planner. Immob. and Energy out are counts of missions ending in embedding and in energy exhaustion respectively.

Condition	Planner	n	Success	Science frac.	Immob.	Energy out
Moon (in-distribution)	Adaptive risk-aware A*	50	0.92	0.829	0	3
	Distance-only A*	50	0.66	0.659	0	12
	Fixed risk-aware A*	50	0.90	0.788	0	4
Mars (OOD)	Adaptive risk-aware A*	50	0.34	0.158	7	20
	Distance-only A*	50	0.48	0.101	7	18
	Fixed risk-aware A*	50	0.34	0.173	6	21
Mars, 1.5x slip dispersion	Adaptive risk-aware A*	50	0.36	0.132	7	22
	Distance-only A*	50	0.44	0.081	8	20
	Fixed risk-aware A*	50	0.22	0.162	8	27
Mars, faults scheduled	Adaptive risk-aware A*	50	0.46	0.146	8	17
	Distance-only A*	50	0.40	0.074	9	20
	Fixed risk-aware A*	50	0.26	0.168	9	24
Mars, 20-step intervention delay	Adaptive risk-aware A*	50	0.44	0.156	2	23
	Distance-only A*	50	0.54	0.107	3	19
	Fixed risk-aware A*	50	0.28	0.200	4	26

Table 2: Primary contrasts: adaptive versus fixed risk-aware A*, paired within condition and seed. Holm-corrected across the primary family; * marks $p_{\text{Holm}} < 0.05$.

Condition	Outcome	Δ 95% CI	Test	p_{Holm}
Mars, 20-step intervention delay	Science fraction	-0.044 [-0.085, -0.002]	paired t-test	0.326
Mars, 20-step intervention delay	Mission success	+0.160 [+0.004, +0.316]	McNemar exact	0.461
Mars, faults scheduled	Science fraction	-0.022 [-0.050, +0.005]	paired t-test	0.461
Mars, faults scheduled	Mission success	+0.200 [+0.085, +0.315]	McNemar exact	0.020*
Mars, 1.5x slip dispersion	Science fraction	-0.030 [-0.059, -0.002]	paired t-test	0.326
Mars, 1.5x slip dispersion	Mission success	+0.140 [-0.001, +0.281]	McNemar exact	0.461
Mars (OOD)	Science fraction	-0.015 [-0.051, +0.022]	paired t-test	1.000
Mars (OOD)	Mission success	+0.000 [-0.128, +0.128]	McNemar exact	1.000
Moon (in-distribution)	Science fraction	+0.040 [+0.002, +0.079]	paired t-test	0.326
Moon (in-distribution)	Mission success	+0.020 [-0.071, +0.111]	McNemar exact	1.000

7 After Study 1: simulator audit and methodological corrections

Everything in this section was found or done after Study 1’s results were seen. None of it changes Study 1’s reported numbers; it changes how some of them should be read, and it shapes Study 2.

Table 3: Generalization gap in science fraction between the in-distribution lunar condition and the out-of-distribution Martian condition. See Section 6 for why this ranking is not informative.

Planner	In-distribution	Out-of-distribution	Gap G
Adaptive risk-aware A*	0.829	0.158	0.670
Distance-only A*	0.659	0.101	0.558
Fixed risk-aware A*	0.788	0.173	0.615

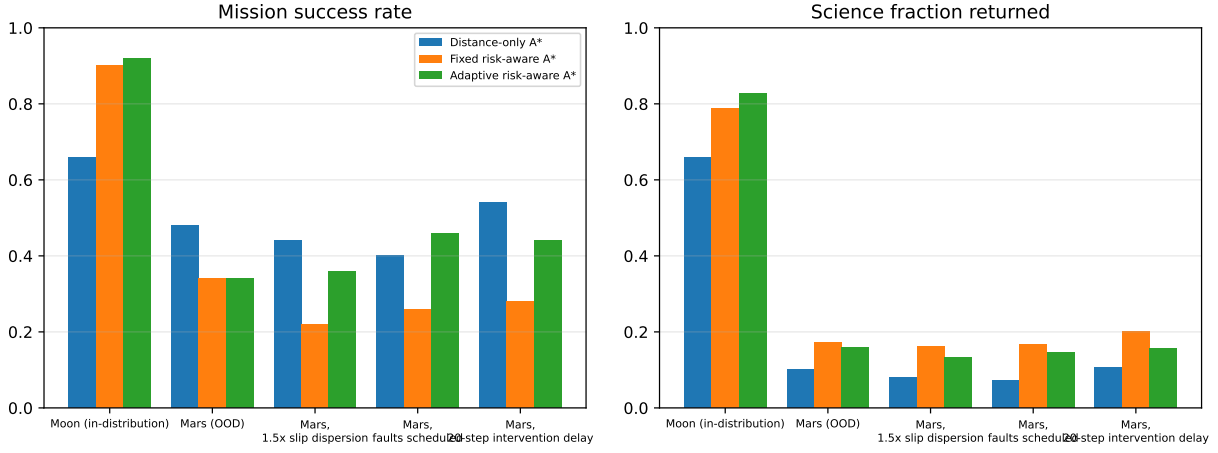


Figure 2: Mission success rate and science fraction by condition and planner.

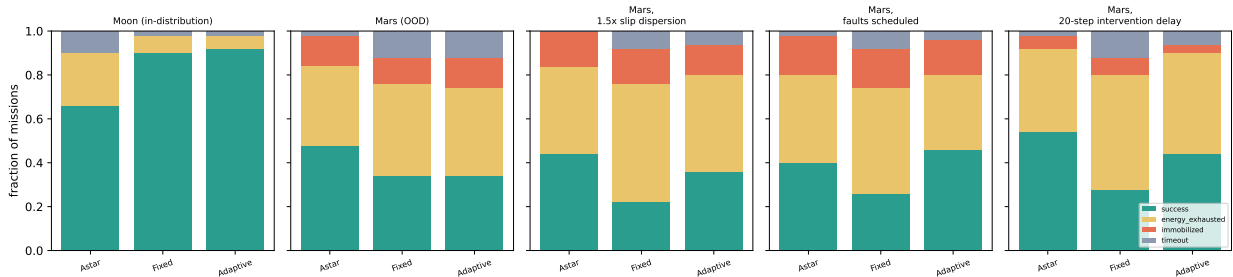


Figure 3: How missions ended. Descriptively, the adaptive planner ends fewer missions in energy exhaustion than the fixed planner in every condition. This was not a pre-specified test.

7.1 Defects and their effect on Study 1

Table 4 lists the defects found. Each is fixed in engine profile *v2*, which is selected explicitly; *v1* remains the default and still reproduces every Study 1 mission on the generating platform.

Defect in v1	Effect on Study 1	Fix in v2
Learner treats each slip reading as a near-exact measurement of the class mean	beliefs overconfident (coverage 0.20)	per-reading variance includes class dispersion
Faults drawn over the whole horizon	“faults” condition mostly fault-free (H3 not a fault effect)	faults inside the mission
One random stream for every purpose	cross-condition comparisons change all noise	separate streams
Hazard-threshold relaxation never reset	hazard avoidance relaxed after early help requests	one planning cycle only
Livelock at the lander	most timeouts are parked rovers	recharge and reconsider
Reported minimum charge can miss the final draw	descriptive <code>min_charge</code> slightly high	final state included

Table 4: Defects found after Study 1. Details and counts: `docs/RESEARCH_LOG.md`.

An exploration of *v2* on validation seeds only (`docs/EXPLORATION_V2.md`, generated from its data) found that the fixes do what they claim, and that with them in place adaptation’s survival advantage on validation seeds largely disappears. It also showed that the communication-delay condition has no mechanism under *v2*, since delay acts only through help requests and *v2* almost never asks for help; Study 2 drops it.

A simulator validation suite now checks terrain, motion, energy accounting, slip sampling, fault timing and streams, planner equivalence, absence of ground-truth access by any planner, and the single enforcement point of the risk budget (`docs/SIMULATOR_VALIDATION.md`). It must pass before Study 2 may run.

7.2 Calibration audit of the terrain learner

Because the planner turns its slip uncertainty into $P(\text{fail})$, the learner’s intervals must mean what they say. On train seeds, the *v2* learner’s nominal 95% interval for a class’s mean slip contained the truth 0.498 of the time on Mars and 0.678 on the Moon — miscalibrated even in distribution. Diagnostic ablations using simulator truth attributed most of this to misclassification: 9.8% (Mars) and 10.3% (Moon) of slip readings were filed under the wrong terrain class, and filing them under the true class alone raised coverage to 0.736 (Mars) and 0.977 (Moon). The rest of the Martian shortfall is the lunar prior’s conflict with Martian loose fines — the domain shift itself.

Six candidate corrections, their fitting data (train seeds only) and an acceptance rule were fixed before any validation run (`docs/CALIBRATION_AUDIT.md`). The candidates were: no change; a confusion-aware learner that shares each reading among the classes that could have produced it, using the rover’s own classifier error rate (no fitted parameter); a conformal scale on the reported uncertainty, fit on lunar or on Martian training missions; and the two combined. The rule required, on Mars, 95% coverage in $[0.90, 0.99]$ and 50, 80 and 90% coverage each within 0.10 of nominal, and on the Moon 95% coverage of at least 0.90.

No candidate met the rule (Figure 4). The confusion-aware learner was calibrated on the Moon at every level (95%: 0.958) but reached only 0.720 on Mars, where the lunar prior’s error on loose fines remains (fines coverage 0.27). The closest scaled candidate, a scale of 4.77 fit on lunar

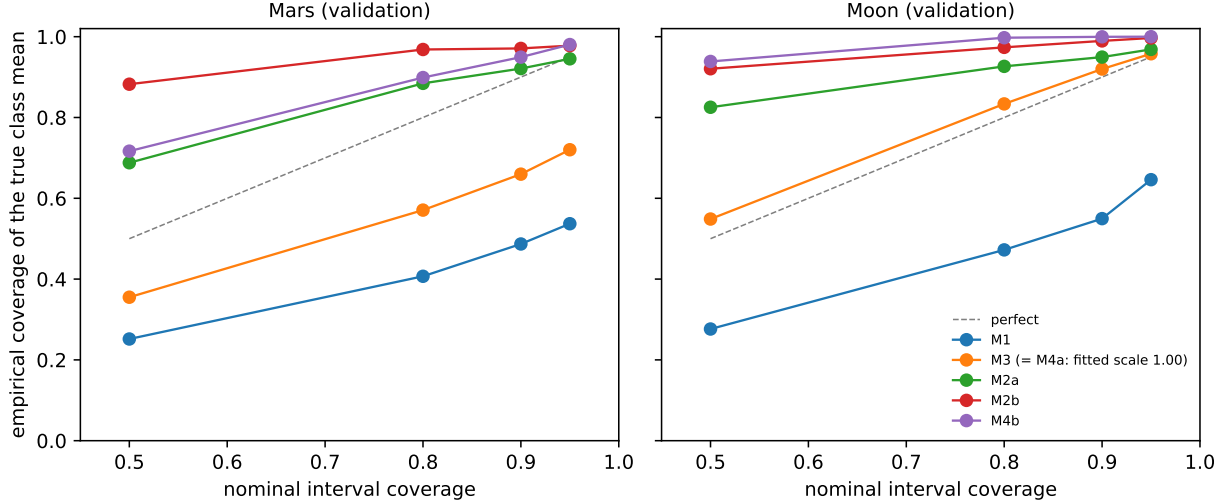


Figure 4: Empirical versus nominal coverage of the class-mean intervals on validation seeds, for each pre-specified calibration candidate. The dashed line is perfect calibration. Development data only; not a confirmatory result.

missions, reached 0.945 at the 95% level on Mars but over-covered at 50% (0.688), because one scale cannot fit a residual distribution in which most beliefs are accurate and a few are far off. The rule was not changed after the result was seen; choosing how to proceed is recorded as an open decision for Study 2.

8 Study 2: frozen, not yet executed

Study 2 has not been run, and this section reports no result. Its analysis plan is frozen in docs/PREREGISTRATION_V2.md; running it requires a separate, explicit authorisation. It asks one narrower question:

Does online adaptation of terrain slip beliefs change safe mission completion under terrain-model shift, compared with an otherwise identical fixed risk-aware planner, and what does it do to science return?

Design. The fixed and adaptive planners differ only in whether measured slip revises the class beliefs. Both use engine v2 (the Study 1 defect fixes and the numerical rules below), the same lunar prior, and the confusion-aware learner. The learner was chosen by the project owner after the calibration rule accepted no candidate (Section 7); its known shortfall on Mars is stated in advance and measured on the confirmatory missions. The confirmatory condition is Mars with a lunar prior; the Moon is secondary; higher dispersion and faults are exploratory; communication delay is dropped. Each condition uses 240 matched seeds from pools disjoint from every Study 1 split.

Hypotheses and tests. Primary: mission success in the Mars condition differs between the planners (two-sided McNemar exact test, $\alpha = 0.05$, one test, no correction), estimated as a difference in proportions with Newcombe’s hybrid score interval. Secondary, Holm-corrected as a family: Mars

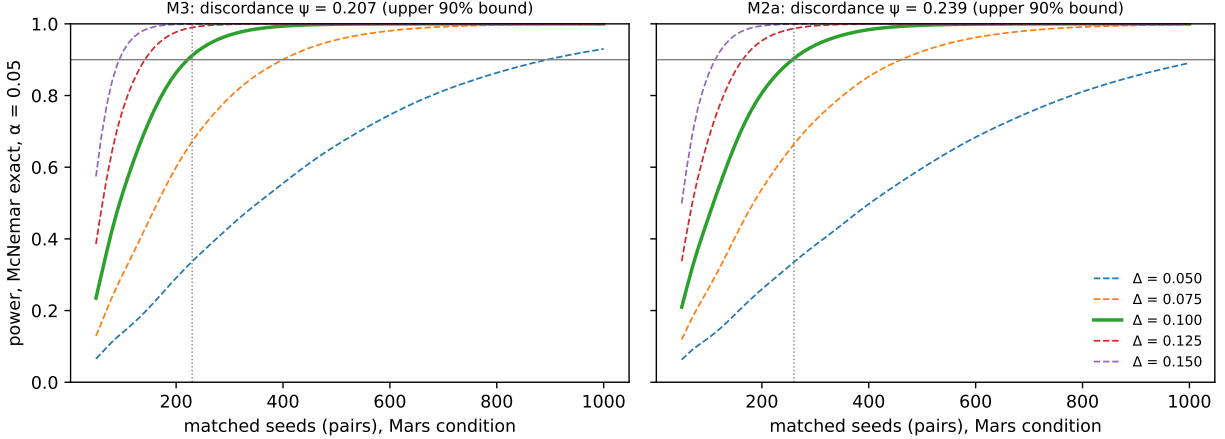


Figure 5: Exact power of the planned primary test against the number of matched seeds, for several effect sizes (solid: the minimum effect of interest), at the upper-bound discordance rate estimated on validation seeds for each leading calibration candidate, with the engine before the numerical changes (the final engine’s rate is slightly higher; see text). Planning quantities only.

science fraction and Moon success and science fraction, all two-sided. No direction is hypothesised: Study 1 found an exact tie in Martian success.

Sample size. The minimum effect of interest is 0.10 in success probability. The sample size is the smallest n , in steps of 10, whose exact power for the primary test reaches 0.90 at that effect, using the upper bound of the planners’ disagreement rate on 200 validation seeds and deliberately ignoring the validation effect itself (Figure 5). Applied to the final engine (24 and 13 seeds where only one planner succeeded; upper bound 0.223), the rule gives $n = 240$ matched seeds per condition, with exact power 0.901. An earlier application, before the numerical changes below, had given 230; with the final engine that n would have had power 0.888, below the target, so the plan was revised to 240 before any confirmatory seed was accessed.

Numerical reproducibility. Because Study 1 reproduces exactly only on its own platform, the Study 2 engine was audited for platform-dependent arithmetic on validation seeds, run on both macOS (arm64) and Linux (x86-64). Continuous terrain values are rounded to six decimals after generation, near-tied target utilities go to the lower target number, and the planner’s exponential, error-function, hypotenuse, integer-power and summation operations use implementations built from correctly rounded IEEE-754 operations, so they give the same bits everywhere. Over the 800 validation missions, the number following the same path on both platforms rose from 661 to 772, success/failure flips fell from 13 to 3, and the Mars effect estimate was +0.055 on macOS and +0.055 on Linux. Terrain became bit-identical on every map and the first decision of every mission agreed exactly, but 192 missions still diverged in the last bits while driving, so exact identity was not reached. By a rule written before the last measurement, macOS is the canonical platform and the Linux re-run is a pre-specified robustness check: the primary conclusion counts as platform-robust if the primary test’s decision matches and the two estimates differ by at most 0.05 (`docs/NUMERICAL_REPRODUCIBILITY.md`).

Protection. The terrain generator refuses every Study 2 confirmatory seed unless the frozen plan exists and the run is authorised with that plan’s hash, and it logs every attempt. At the time of writing no Study 2 confirmatory terrain has been generated.

9 Discussion

The result we set out to find is not the result we obtained. H2, the hypothesis that online adaptation would raise science return under domain shift, is not supported: the effect was absent on held-out seeds despite appearing clearly during tuning. The one contrast that survived correction turned out, on audit, not to measure what its condition was named for, so H3 is not supported either. What remains is a descriptive pattern in which adaptation trades a little science for survival. It is consistent with a plausible mechanism, but it was not tested as such, and it comes from a learner now known to be miscalibrated.

Three things follow.

First, the interesting version of this experiment has not been run yet. Study 2 is designed to be that version, and preparing it has already produced a finding of its own: the learner at the centre of the adaptive planner was miscalibrated even on familiar terrain, mostly through terrain misclassification, and while that can be fixed without fitting anything, no simple correction calibrated it under the domain shift itself. A risk-aware planner inherits whatever its uncertainty model gets wrong, so this is not a side issue for Study 2.

Second, caution has a price that risk-aware formulations do not currently charge. On terrain where most ground is hazardous, detouring is not free, and the uninformed planner’s survival advantage came from simply spending less time exposed. A more complete objective would price route length and exposure time against per-cell risk rather than treating risk avoidance as costless.

Third, the gap between the validation and confirmatory estimates is the most methodologically useful outcome here. A tuning-set advantage of +0.125 to +0.167 in Martian success became 0.000 on held-out seeds. Had the seed protocol not been frozen in advance, that first number is the one that would have been reported.

9.1 Future work

The obvious extensions are to adapt the locomotion multiplier as well as the slip model, to add exposure time to the planner’s objective so that caution is priced, and to test transfer in the reverse direction (Mars prior, lunar deployment) to check that the effects here are not specific to one direction of shift.

10 Limitations

Stated plainly, because the value of this study depends on what it does not claim.

1. **This is not a validated rover model.** Terrain parameters were chosen to produce the qualitative regimes the experiment needs — a low-traction trap, a high-traction refuge, impassable rims, and true darkness — and are not calibrated against measured lunar or Martian geotechnical data. Absolute values are properties of this simulator.
2. **2.5-D, not dynamics.** There is no vehicle suspension, no wheel contact mechanics, and no attitude. Slip is drawn from a distribution rather than emerging from terramechanics.

3. **Risk composition is approximate.** Equations 6 and 7 treat per-cell events as independent, which understates risk in homogeneous terrain. It is applied identically to all planners.
4. **Only the slip model adapts.** The per-class locomotion multiplier is not revised online. Energy estimates still improve through the slip channel, since slip enters Equation 2 directly, but a systematic error in m_k would persist.
5. **The analysis plan is not a true pre-registration.** It was written after the adaptive planner existed and after an eight-seed engineering pilot had been seen. Those eight seeds are quarantined, and the full timeline is recorded in `docs/RESEARCH_LOG.md`.
6. **Two bodies, one prior direction.** Only Moon→Mars transfer is tested. The reverse direction, and bodies beyond these two, are untested.
7. **The adaptive learner is overconfident.** The conjugate update (Equation 3) uses an observation variance of 0.02^2 per slip reading. That treats a reading as a near-exact measurement of the class mean, when readings actually scatter by the terrain’s own dispersion (standard deviation 0.04–0.18). One reading therefore moves the belief most of the way to itself, and the reported uncertainty is far too small (95% coverage 0.20; see Results). All confirmatory results use this rule.
8. **Faults mostly did not fire.** Fault times are drawn over the whole 800-step horizon, but missions end much sooner, so the “hardware faults” condition exposed only 16%–26% of missions to any fault. It is not a test of fault tolerance.
9. **One random stream serves every purpose.** Terrain, layout, fault schedule and every slip draw come from one generator in sequence. With no faults configured, the fault draw is skipped, so every later slip draw shifts. Paired contrasts within a condition are unaffected, because both planners in a pair share a stream. Comparisons across conditions, however, change the whole noise realisation, not just the named factor.
10. **Intervention relaxation ratchets.** Each ground intervention raises the planner’s hazard threshold by 0.15, up to 0.95, and it is never restored. Martian missions requested many interventions, so in most of them the planners’ hazard avoidance was effectively relaxed early. This applies to every planner equally.
11. **Most timeouts are a livelock at the lander.** When the rover is home and no target fits the risk budget, the mission logic neither writes the remaining targets off nor stops. It keeps requesting interventions until the step limit, so most “timeout” failures are rovers parked safely at the lander. Planners reach this state at different rates, so it affects the success comparison; the counts are in `docs/RESEARCH_LOG.md`.
12. **The five items above, from the learner’s overconfidence to the livelock, are Study 1 defects,** fixed in engine profile v2 (Section 7); Study 1’s numbers are reported as run.
13. **Calibration under shift is unsolved.** On development data no pre-specified correction calibrated the learner on Mars at every level (Section 7).
14. **Idealisations in the learner.** It removes the slope’s effect on a reading using the true slope of the cell driven, standing in for a noise-free attitude sensor; slip is spatially independent between cells.

11 Reproducibility

Every reported number is a macro generated from a committed artifact; no result value is typed into this document. `paper/PROVENANCE.yaml` records, for every generated table, figure and macro, the script that produced it, its hash, its input files and their hashes, and the commit; a test fails if any of them is stale or edited by hand, and continuous integration re-runs every generator.

Cross-platform reproduction. Study 1 reproduces exactly on the environment that generated it (macOS, Apple Silicon/arm64, with the pinned libraries); this was re-checked after all later engine changes. Re-run on Linux x86-64 with the same code and pinned libraries, 672 of 750 missions are identical and 78 differ: last-bit floating-point differences between platforms change a route choice in some missions, and 18 change outcome between success and failure. Recomputing the pre-specified analysis on the Linux outcomes moves no paired effect by more than 0.020 and leaves every significance verdict unchanged (same verdicts: yes); the one significant contrast is +0.22 with Holm $p = 0.010$ on Linux. The conclusions are therefore not platform artefacts, but exact reproduction is a property of the reference platform only, and the tests that demand it say so. The run’s metadata sidecar records the git commit, the seed-split checksum, the quarantine list and library versions. The pipeline is:

```
python scripts/freeze_seed_protocol.py      # once; refuses to overwrite
python scripts/calibrate_priors.py          # priors from TRAIN seeds only
python scripts/run_exonaut_experiments.py   # Study 1 confirmatory sweep
python scripts/verify_v1_reproduction.py    # all 750 Study 1 missions
python scripts/calibration_study.py diagnose # then fit, evaluate, report
python scripts/power_analysis.py run        # then analyze
python scripts/build_all_figures.py         # every figure, table, macro
```

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